PHYSICAL QUANTITIES

Physical Quantities and Units

The field of physics is divided into the areas of mechanics, waves, optics, thermodynamics, and electricity which make up the classical physics developed between the sixteenth and nineteenth centuries Galileo, Newton, Faraday, Maxwell, and others. Physics also includes modern topics developed in the twentieth century, such as relativity, quantum physics, and quantum field theory, as well as their appropriate applications in condensed matter, atomic and nuclear phenomenon, elementary particles, and astrophysics. Physics is sometimes defined as the study of energy and matter and their relationships. Physics is also inherently a science of measurements and the ultimate test for any theory in physics is its agreement with observations and the measurements of the physical phenomenon. Thus, any number or set of numbers used for the quantitative description of a physical phenomenon is called a physical quantity. To define a physical quantity we must have a procedure for measuring such quantity or specify a way of calculating it. We measure a physical quantity in its own units and by comparison with a standard unit.

SI standard units

By international agreement a small number of physical quantities such as length, time and mass (called Fundamental quantities) have been assigned standard units. We can define all other physical quantities in terms of these fundamental quantities. The units of these fundamental quantities are standard units (Table 1.1) and are both accessible and invariable.

Table 1.1 SI units for fundamental quantities

Quantity	Unit Name	Unit Symbol
Length	meter	m
Time	second	s
Mass	kilogram	kg
Temperature	kelvin	K
Electric Charge	Coulomb	C
Amount of a substance	mole	mol

Derived Units

Table 1.2 SI units for derived quantities

Quantity	Unit Name	Unit Symbol
Angle	radian	rad
Solid angle	steradian	sr
Area	square meter	m^2
Volume	cubic meter	m^3
Density	kilogram per cubic meter	kg/m^3
Velocity	meter per second	m/s
Acceleration	meter per second squared	m/s^2
Angular velocity	radian per second	rad/s
Angular acceleration	radian per second squared	rad/s^2
Force	Newton	N ($kg\ m/s^2$)
Torque	newton-meter	Nm
Work, Energy, heat	joule	J (N m)
Power, radiant flux	watt	W
Pressure, stress	pascal	Pa (N/m^2)
Frequency	hertz	Hz (1/s)

Electric current	ampere	A
Voltage	Volt	V (W/A)
Electric field	volt per meter	V/m
Electrical resistance	ohm	Ω (V/A)
Capacitance	farad	F (A s/V)
Magnetic flux	weber	Wb (V s)
Inductance	henry	H (V s/A)
Magnetic Flux density	tesla	T (Wb/m^2)
Luminous flux	lumen	Lm
Luminance	lux	lx
Activity (radioactive)	becquerel	Bq (s^{-1})
Absorbed dose	gray	Gy (J/kg)
Dose equivalent	sievert	Sv (J/kg)

Standard Prefixes

In physics we sometimes deal with numbers that are very small and other times we deal with numbers which are very large. It gets cumbersome to write numbers in the conventional decimal notation. Once the fundamental units are defined, it now becomes easy to introduce larger and smaller units for the same physical quantities. In the metric system measurements additional units always relate to the fundamental units by multiples of 10 or $\frac{1}{10}$. The multiplicative factors are most conveniently expressed in exponential notation; thus $10000 = 10^4, \frac{1}{100} = 10^{-2}$ etc.

The names of the additional units are always derived by adding a prefix to the name of the fundamental units.

Table 1.3 Prefixes for the power of ten

Power of ten	Prefix	Abbreviation
10^{-24}	yocto-	y
10^{-21}	zepto-	z
10^{-18}	atto-	a
10^{-15}	fermto-	f
10^{-12}	pico-	p
10^{-9}	nano-	n
10^{-6}	micro-	μ
10^{-3}	milli-	m
10^{-2}	centi-	c
10^{3}	kilo-	k
10^{6}	mega-	M
10^{9}	giga-	G
10^{12}	tera-	T
10^{15}	peta-	P
10^{18}	exa-	E
10^{21}	zetta-	Z
10^{24}	yotta-	Y

The following are a few examples of the use of the multiples of ten and their prefixes.

1 nanometer = 1nm = 10^{-9} m

1 micrometer = 1μm = 10^{-6} m

1 millimeter = 1mm = 10^{-3} m

1 centimeter = 1cm = 10^{-2} m

1 kilometer = 1km = 10^{3} m

1 milligram = 1mg = 10^{-6}kg

1 gram = 1g = 10^{-3} kg

1 nanosecond = 1ns = 10^{-9} s

1 microsecond = 1ms = 10^{-3} s

Unit Conversions

Relations among physical quantities are often expressed by equations in which physical quantities are represented by algebraic symbols. An algebraic symbol always denotes both a number and a unit. For example, *x* might represent distance of 5m, *t* a time of 30s and *m* a mass of 10kg. We often need to convert the units in which a physical quantity is expressed. We do so by multiplying the original measurement by a conversion factor which is a ratio of the units in question that equals unity. For example

$$\frac{1\min}{60s} = 1 \text{ and } \frac{60s}{1\min} = 1$$

This is so because multiplying any physical quantity (or algebraic symbol) by unity leaves the quantity unchanged. For example we can convert 5min into seconds by multiplying it by the conversion factor of $\frac{60s}{1\min}$ to have

$$5\min = (5\min)(1) = (5\min)(\frac{60s}{1\min}) = 300s$$

Dimensions

The fundamental quantities used in physical descriptions are called dimensions. For example length, time and mass are dimensions. You could measure the distance between two points and express it in units of meters, centimeters, or feet, but the quantity would still have the dimension of length. It is common to express dimensional quantities by bracketed symbols as [L], [M] and [T] for length, mass and time respectively. Derived quantities are a combination of dimensions. For example, velocity v has dimensions of $[L]/[T]$ (think of it as meters/second = length/time) and volume, V has dimensions $[L]\times[L]\times[L]$, or $[L^3]$.

Exercise 1.1 What is dimension (or the dimension of a physical quantity)? What is the dimension of acceleration?

Solution:

The dimensions of a physical quantity are the powers (or exponents) to which the base or the fundamental quantities are raised to represent that quantity. For example, acceleration can be expressed as

$$acceleration = \frac{length}{(time)^2} = \frac{[length]}{[time]^2} = \frac{L}{T^2} = LT^{-2}$$

The dimensions of acceleration are then given by

$$[acceleration] = [LT^{-2}]$$

Thus, we conclude that, acceleration has no or zero dimension in mass, one dimension in length, and -2 dimensions in time. Any quantity raised to zero is equal to 1. So, all those fundamental quantities that do not appear in the dimensional formula of acceleration have dimension 0. For example, here, the dimension of mass is 0.

Dimensions and unit analysis

Any equation must always be dimensionally consistent or homogenous-this means that any two terms added or subtracted in an equation must have the same units. For example given the equation: 3.0m + 2.0m = 5.0m (similar to $a + b = c$), comparing both sides we have $[L] + [L] = [L]$ or in words length plus length gives you another length not 2[L]. The above procedure for checking units in equations is called dimensional analysis. It is formally defined as a procedure by which the dimensional consistency of any equation may be checked. One major advantage of dimensional analysis is its usefulness for checking whether an equation that has been derived or being used to solve a problem has the correct form. The dimensions and units of some common quantities are given in Table 1.3.

Exercise 1.2 What are dimensional equations?

Solution:

Dimensional equations are equations, which represent the dimensions of a physical quantity in terms of the fundamental quantities. When one equates a physical quantity with its dimensional formula then we have a dimensional equation of the physical quantity. The dimensional equations can be obtained from the equations representing the relations between the physical quantities. This is something that we have been doing, writing the quantity on the left hand side of the equation and its dimensions on the right hand side.

Example 1.3 In the gas equation $(p+\frac{a}{V^2})(V-b)=RT$, what are the dimensions of the constants a and b?

Solution:

The dimensions of the terms in the equation are as follows:

$$[p]=[\text{Pressure}]=\frac{[\text{Force}]}{[\text{Area}]}=\frac{MLT^{-2}}{L^2}=ML^{-1}T^{-2}$$

$$[V]=[volume]=L^3$$

Since $\frac{a}{V^2}$ is added to pressure and we assume the given equation to be homogenous it must be pressure.

$$\Rightarrow\left[\frac{a}{V^2}\right]=\frac{[a]}{[V^2]}=\frac{[a]}{L^6}=[p]=ML^{-1}T^{-2}$$

$$\therefore[a]=ML^{-1}T^{-2}\times L^6=ML^5T^{-2}$$

The constant b must represent a volume since it is being subtracted from a volume

$\Rightarrow [b] = L^3$

Example 1.4 By using dimensional analysis, derive an expression for the volume V of water passing any section of a pipe in time (i.e. the rate of volume flow V/t) if it depends on (a) the coefficient of viscosity η of the liquid (Note: $[\eta] = ML^{-1}T^{-1}$ (b) the radius r of the pipe (c) the pressure gradient $\Delta p/l$, where Δp is the pressure difference between the ends of the pipe and l its length.

Solution:

We express the relations of the flow rate to the viscosity, radius and pressure gradient as:

$$\frac{V}{t} = k\eta^x r^y \left(\frac{\Delta p}{l}\right)^z$$

Where k is a dimensionless constant and x, y and z are unknown indices. Since each side of the equation must have the same dimensions we can write

$$\left[\frac{V}{t}\right] = [\eta^x][r^y]\left[\left(\frac{\Delta p}{l}\right)^z\right]$$

$$\therefore L^3T^{-1} = (ML^{-1}T^{-1})^x (L)^y (ML^{-2}T^{-2})^z$$

$$\Rightarrow L^3T^{-1} = M^x L^{-x} T^{-x} \cdot L^y \cdot M^z L^{-2z} T^{-2z}$$

$$\Rightarrow M^0 L^3 T^{-1} = M^{x+z} L^{y-x-2z} T^{-x-2z}$$

Equating the indices of M, L and T on both sides

For M: $0 = x + z$ (1)

For L: $3 = y - x - 2z$ (2)

For T: $-1 = -x - 2z$ (3)

Solving equations 1, 2 and 3 simultaneously gives $x = -1$, $z = 1$ and $y = 4$. The relationship is therefore

$$\frac{V}{t} = k\frac{r^4}{\eta}\left(\frac{\Delta p}{l}\right)$$

This relation is called Poiseuille's Law or relation. (note that $k = \pi/8$)

Example 1.5 Consider a sphere of radius r moving with velocity v through a fluid whose coefficient of viscosity is η. The sphere experiences a viscous force F which acts in the opposite direction to that in which the sphere is moving. Use dimensional analysis to obtain an expression for $F \propto r\eta v$.

Solution:

$F = kr^x\eta^y v^z$

$[F] = [r^x][\eta^y][v^z]$

$MLT^{-2} = L^x(ML^{-1}T^{-1})^y(LT^{-1})^z$

$M^1L^1T^{-2} = M^yL^{x-y+z}T^{-y-z}$

For M: $1 = y$ (1)

For L: $1 = x - y + z$ (2)

For T: $-2 = -y - z$ (3)

Solving equations 1, 2 and 3 simultaneously gives $y = 1$, $z = 1$ and $x = 1$. Thus

$F = kr\eta v$

Note that $k = 6\pi$.

Example 1.6 Check if the following equations are dimensionally correct:

(a) $x = at$ (b) $x = at^2$, where x is distance, a is acceleration and t is time.

Solution:

(a) Equation $x = at$ can be expressed dimensionally as

$[L] = \frac{[L]}{[T^2]} \times [T] = \frac{[L]}{[T]}$ which is not true. Thus $x = at$ cannot be a correct equation.

(b) Equation $x = at^2$ can be expressed dimensionally as

$[L] = \frac{[L]}{[T^2]} \times [T^2] = [L]$. This equation is dimensionally correct but as you will see later it is not always physically correct. The correct form of $x = at^2$ both dimensionally and physically is $x = \frac{1}{2}at^2$. The fraction $\frac{1}{2}$ has no dimensions i.e. it is a dimensionless number. Doing dimensional analysis with the symbols is fine but in practice it is often more convenient to use actual unit abbreviations such as m, s, and kg. Units can also be treated as symbols and canceled. Using units instead of symbols in dimensional analysis is called units analysis.

Exercise 1.7 Is the equation $s = ut + 1/2\,at^2$ dimensionally consistent?

Solution:

We are given $s = ut + 1/2\,at^2$ or

s(displacement) = u(initial velocity) X t(time) + 1/2 X a(acceleration)X t^2(time)

We solve by going through the following steps:

Step 1: Identify the terms in the equation and identify which symbol stands for which physical quantity.

There are three terms here: s, ut, at^2

Step 2: Write down the dimensional formula of each symbol used in the equation

[s] = [L], [u] = $[LT^{-1}]$, [t] = [T], [a] = $[LT^{-2}]$

Step 3: Calculate the dimension of each term in the equation.

Term 1: [s] = [L]

Term 2: [ut] = $[LT^{-1}.T]$ = [L]

Term 3: $[at^2]$ = $[LT^{-2}.T^2]$ = [L]

Note that, as stated above, we have canceled dimensions from numerator and denominator like $[T^{-2}.T^2]$.

Conclusion: The equation is dimensionally consistent since all the terms have the same dimensions.

Note: The following kinds of terms are dimensionless in an equation.

- A pure number or the ratio of similar physical quantities, such as angle as the ratio (length/length), refractive index as the ratio (speed of light in vacuum/speed of light in medium) etc are dimensionless.
- The arguments of special functions, such as the trigonometric, logarithmic and exponential functions are also dimensionless.

Example 1.8 Check the dimensional consistency of the following equations:

(a) $K = (1/2)mv^2 + ma$ (b) $K = (1/2)mv^2$ (c) $K = (3/16)mv^2$

Solution:

(a) The first term (on the LHS),

$K = Kinetic\ Energy = Work = \text{Force} \times \text{Displacement}$

$\Rightarrow [K] = [MLT^{-2}] \times [L] = [ML^2T^{-2}]$

The second term involves, m = Mass and v = velocity; and it has dimensions:

$[(1/2)mv^2] = [1/2][m] \cdot [v^2] = [m] \cdot [v]^2 = [M] \cdot [LT^{-1}]^2 = [M] \cdot [L^2T^{-2}] = [ML^2T^{-2}]$

The third term involves, m = Mass and a = acceleration

$[ma] = [m][a] = [M][LT^{-2}] = [MLT^{-2}]$

It can be recognized that the dimensions of the first two terms have the same dimension but that of the third term differs. Therefore the equation $K = (1/2)mv^2 + ma$ is dimensionally inconsistent.

(b) The first term (on the LHS),

$K = \text{Kinetic Energy} = \text{Work} = \text{Force} \times \text{Displacement}$

$\Rightarrow [K] = [MLT^{-2}] \times [L] = [ML^2T^{-2}]$

The second term involves, m = Mass and v = velocity; and it has dimensions:

$[(1/2)mv^2] = [1/2][m] \cdot [v^2] = [m] \cdot [v]^2 = [M] \cdot [LT^{-1}]^2 = [M] \cdot [L^2T^{-2}] = [ML^2T^{-2}]$

Since the dimensions of the two terms of the equation are dimensionally similar then the equation $K = (1/2)mv^2$ is dimensionally consistent.

(c) Just like in **(a)** and **(b)**, the first term (on the LHS),

$K = Kinetic\ Energy = Work = Force \times Distance$

$\Rightarrow [K] = [MLT^{-2}] \times [L] = [ML^2T^{-2}]$

The second term involves, m = Mass and v = velocity; and it has dimensions:

$$[(3/16)mv^2] = [3/16][m]\cdot[v^2] = [m]\cdot[v]^2 = [M]\cdot[LT^{-1}]^2 = [M]\cdot[L^2T^{-2}] = [ML^2T^{-2}]$$

Since the dimensions of the two terms of the equation are dimensionally similar then the equation $K = (3/16)mv^2$ is dimensionally consistent. Note that the dimension of a number, an argument or a ratio is unity.

Table 1.4 Dimensions and Units

Quantity	Dimension	Unit
mass	$[M]$	kg
time	$[T]$	s
length	$[L]$	m
area	$[L^2]$	m^2
volume	$[L^3]$	m^3
velocity	$[L]/[T]$	m/s
Acceleration (a or g)	$[L]/[T^2]$	m/s^2

Precision and Accuracy

Accuracy is a measure of how close a result of an experimental observation comes to the "true" value. Precision on the other hand is a measure of how carefully a result is determined without reference to any "true" value. Measurements can never be made with absolute precision in that physical quantities obtained from experimental observations always have some uncertainty. A distance measured with an ordinary ruler or meter rule is usually precise only to the nearest millimeter, whilst a precision micrometer caliper can measure a distance to the nearest 0.01mm

or even less. The precision of a measurement or number is often indicated by following the number with the symbol ± and a second number indicating the maximum likely error. Errors are defined as the difference between measured experimental values and the "true" value. It is generally applied to the uncertainty in measurements but not blunders or mistakes. If for example, the diameter of a metal rod is given as 48.64±0.01mm, then it means that the true value is likely to be less than 48.65 or greater than 48.63. Precision can also be expressed in terms of the maximum likely fractional error or percentage error. A resistor labeled 47Ω, 10% probably has true resistance differing from 47Ω by no more than 10% of 47Ω or about 5Ω i.e. the true resistance is between 42Ω and 52Ω.

Significant Figures

If a student is asked to find the numerical value of π by using a calculator, after measuring the diameter and circumference of a circle as 135mm and 424mm respectively, pi is the ratio of the circumference to the diameter of a circle. They get an answer like 3.140740741…… Depending on the type of calculator, they could get decimal places more than the number stated. Does this number or any other agree with the true value of π (correct value to ten digits is 3.141592654) or not is the question. To answer this question we must recognize that at least the last six digits in the student's calculation are meaningless unless they imply a greater precision in the result than possible with their measurement of π. The number of meaningful digits in a number is called the number of significant figures. In general, no numerical result can have more significant figures than do numbers from which it was computed. Thus the student's value of π has only three significant figures

and should be stated as 3.14 or (3.141 when rounded of to four figures). The rules for significant figures and rounding offs are as follows:

Rules for Significant Figures:

1. The leftmost nonzero digit is the most significant digit/figure.
2. If there is no decimal point, the rightmost nonzero digit is the least significant digit/figure.
3. If there is a decimal point, the rightmost digit is the least significant digit/figure, even if it is zero.
4. All digits between the least and the most significant digits are counted as the significant figures.

Round off:

Rounding off is simply to truncate a number to the specified number of significant figures and treat the excess (remainder) digits as a decimal fraction.

1. If the fraction is equal to or greater than $1/2$ (0.5) then increase the new least significant figure by one.
2. If the fraction is less than $1/2$ (0.5) then leave the new least significant figure unchanged.

Rules for Reporting with the proper Significant Figures.

Addition/Subtraction

When adding or subtracting numbers, leave the same number of decimal places in the answer as there is in the number with the least number of significant figures.

- For example, suppose we need to add 701.3, 104.0, 41.216 and 0.5462. The result from the calculator is 847.0622. Now, look at the data and observe that in the two of the numbers being added it is not clear what the figure in the second decimal place is. The result must be rounded off to one decimal place by dropping 622. Hence, the answer with proper numbers of significant is 847.0.

Multiplication/Division

When multiplying and dividing numbers, leave as many significant figures in the answer as there are in the number with the fewest number of significant figures. Usually, multiplication and division produce more figures that are significant thus, the result must be rounded off.

- For example, suppose the measured length and width of a plate is 2.64cm and 1.18cm respectively. The area therefore equals $2.64cm \times 1.16cm = 3.1152cm^2$. The result appears to have five significant figures, although the original quantities have three each. Therefore, the result can only have three significant figures and is $3.12cm^2$ (rounded off to three significant figures i.e. rounded to the second decimal place)

Example 1.9 Express the following numbers in scientific notation with three significant figures: (a) 0.00241 (b) 79,241 (c) 0.000452.

Solution:

The numbers are expressed in scientific notation with three significant as:

(a) $0.00241 = 2.41 \times 10^{-3}$

(b) $79{,}241 = 7.92 \times 10^4$ [Note: 7.9241 has been rounded off to 7.92]

(c) $0.0000452 = 4.52 \times 10^{-5}$

Example 1.10 Express the following numbers as their decimal equivalents:

(a) 49.6×10^{-2} (b) -6.50×10^{-4} (c) 0.4291×10^{3} (d) 1.21×10^{0}

Solution:

The decimal equivalents are as follows:

(a) $49.6 \times 10^{-2} = 0.496$ [by moving the decimal point two places left]

(b) $-6.50 \times 10^{-4} =$ -0.000650 [by moving the decimal point four places left]

(c) $0.4291 \times 10^{3} = 429.1$ [by moving the decimal point three places right]

(d) $1.21 \times 10^{0} = 12.1$

Example 1.11 Express the final result of the following expressions in scientific notation: (a) 51.5×0.0000521 (b) $\dfrac{2348}{83.1 - 91.8}$.

Solution:

(a) 51.5×0.0000521

First, express each term in the expression in scientific notation before final computation.

51.5×0.0000521

$= (5.15 \times 10^{1}) \times (5.21 \times 10^{-5})$

$= (5.15 \times 5.21) \times 10^{-4} = 26.8315 \times 10^{-4}$

The result is not correctly nwritten in scientific notation. To get the decimal place in the right place , we move one decimal place to the left, increasing the exponent by +1. Thus, the final answer in scientific notation is 2.68315×10^{-3}. Keeping the

results up to three significant figures, we round off the result and rewrite it as 2.68×10^{-3}.

(b) We first express each number in the fraction in scientific notation.

$$\frac{2348}{83.1-91.8}=\frac{2.348\times10^{3}}{8.31\times10^{1}-9.18\times10^{1}}=\frac{2.348\times10^{3}}{-0.87\times10^{1}}=-2.70\times10^{2}$$

Here the result is shown in scientific notation up to three significant figures.

Equations and Greek symbols

Different symbols and letters are used when stating and solving equations in physics. Table 1.5 shows symbols used in writing physical equations and Table 1.6 shows Greek letters are often used to represent physical quantities in equations.

Table 1.5 Symbols and letters in physical equations

SYMBOL	MEANING	SYMBOL	MEANING
$=$	Equal to	$\therefore$	Therefore
$\approx$	Approximately equal to	Σ	Sum
$\sim$	Same order as	Π	Product
$>$	Greater than	$\sqrt{\ }$	Square root
$\geq$	Greater than or equal to	$x!$	Factorial of x
$<$	Less than	$\neq$	Not equal to
$\leq$	Less than or equal to	∞	Infinity
$>>$	Much greater than	$\propto$	Proportional to
$<<$	Much less than	$\pm$	Plus or minus
$\equiv$	Equivalent to	$\mp$	Minus or plus
$\Rightarrow$	Implies	Δx	Change in x
/	Divided by		

Table 1.6 Greek letters

Letter	Uppercase	Lowercase	Letter	Uppercase	Lowercase
Alpha	Α	α	Nu	Ν	ν
Beta	Β	β	Xi	Ξ	ξ
Gamma	Γ	γ	Omicron	Ο	o
Delta	Δ	δ	Pi	Π	π
Epsilon	E	ε	Rho	Ρ	ρ
Zeta	Ζ	ς	Sigma	Σ	σ
Eta	Η	η	Tau	Γ	τ
Theta	Θ	θ	Upsilon	Υ	υ
Iota	Ι	ι	Phi	Φ	ϕ
Kappa	Κ	κ	Chi	Χ	χ
Lambda	Λ	λ	Psi	Ψ	ψ
Mu	Μ	μ	Omega	Ω	ω